10 Zac is measuring the growth of a culture of bacteria in a laboratory. The initial area of the culture is 8 cm^2. The area one day later is 8.8 cm^2.

At first, Zac uses a model of the form $A = a + bt$, where $A\text{ cm}^2$ is the area t days after he begins measuring and a and b are constants.

(a) Find the values of a and b that best model the initial area and the area one day later. **[2]**

(b) Calculate the value of t for which the model predicts an area of 15 cm^2. **[1]**

(c) Zac notices the area covered by the culture increases by 10% each day.

Explain why this model may not be suitable after the first day. **[1]**

Zac decides to use a different model for A. His new model is $A = Pe^{kt}$, where P and k are constants.

(d) Find the values of P and k that best model the initial area and the area one day later. **[3]**

(e) Calculate the value of t for which the area reaches 15 cm^2 according to this model. **[2]**

(f) Explain why this model may not be suitable for large values of t. **[1]**

10(a)

$a =$

$b =$

10(b)

10(c)

10(d)

$P =$

$k =$

10(e)

10(f)